Crop Yield Prediction Using Machine Learning

An end-to-end machine learning pipeline to predict crop yield.

Utilizes soil nutrients and environmental factors for accurate estimates.

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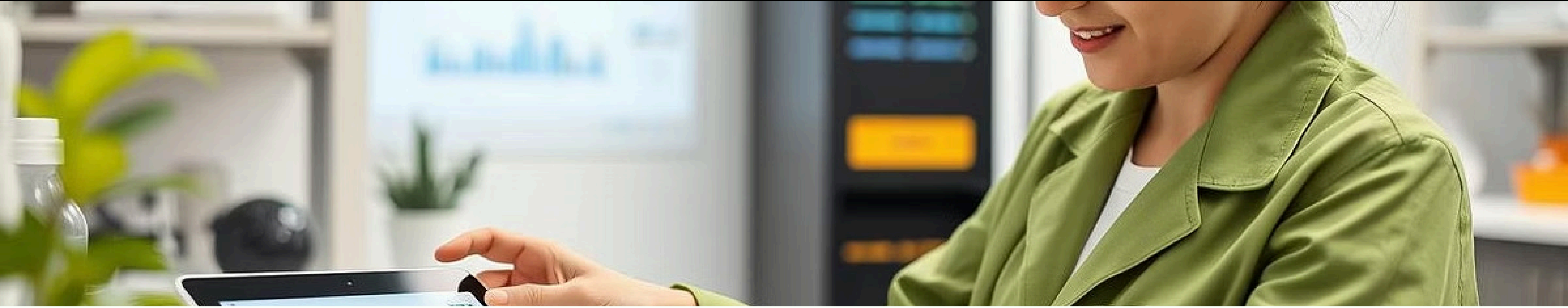


Project Introduction

Objective

Predict crop yield using soil nutrients and environment data.

Features

- Nitrogen, Phosphorus, Potassium
- Temperature, Humidity, pH, Rainfall

Importance

Helps optimize farming practices and improve food production.

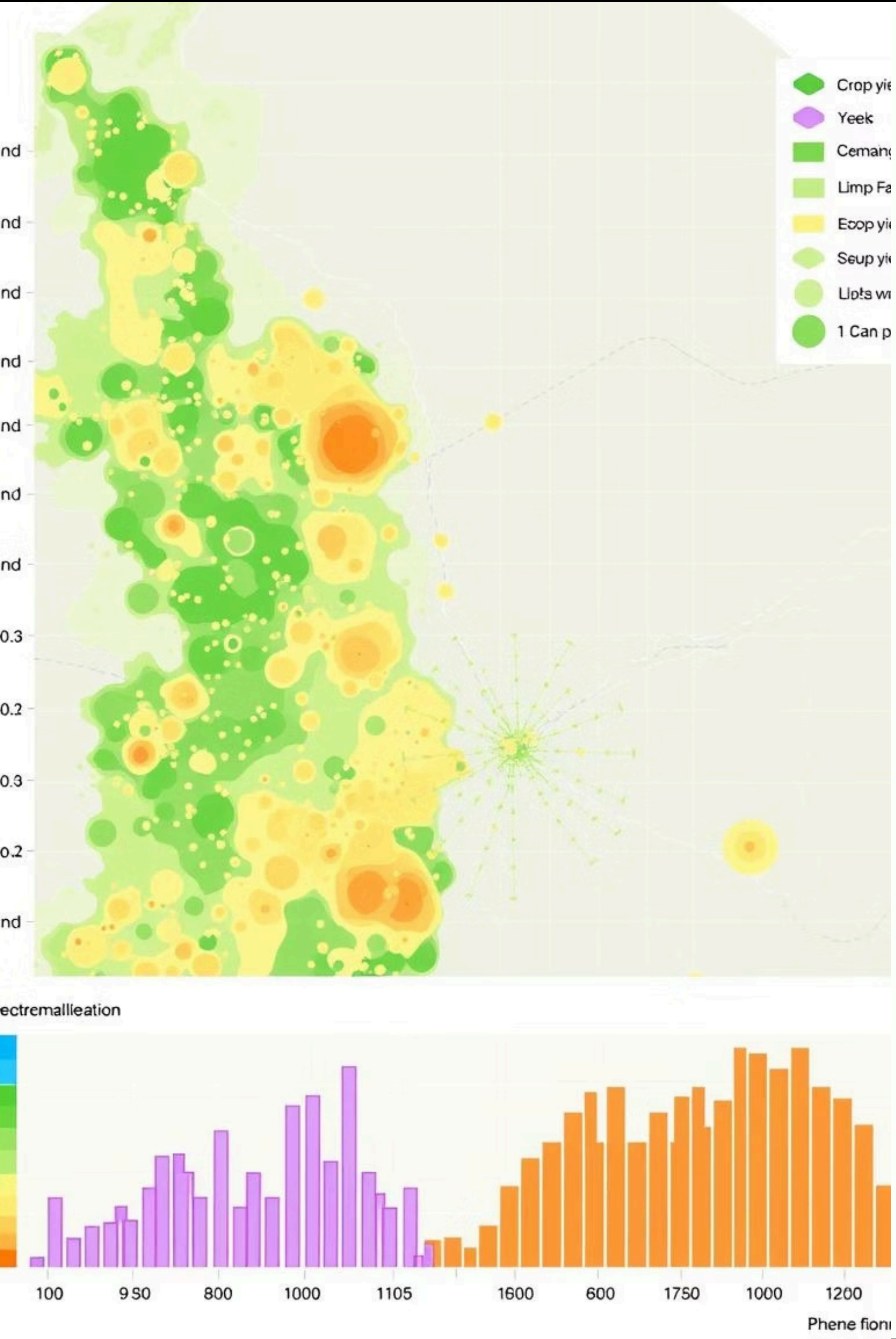
Dataset Overview

Soil Nutrients

- Nitrogen (N)
- Phosphorus (P)
- Potassium (K)

Environmental Factors

- Temperature (°C)
- Humidity (%)
- pH level
- Rainfall (mm)



Exploratory Data Analysis

Feature Distributions

Analyzed nutrient and environmental variable ranges.

Correlations

Identified strong relationships among features and yield.

Visual Insights

Heatmaps and histograms helped detect patterns and outliers.

Data Preprocessing Steps

1

Missing Values

Detected and imputed missing nutrient and environmental entries.

2

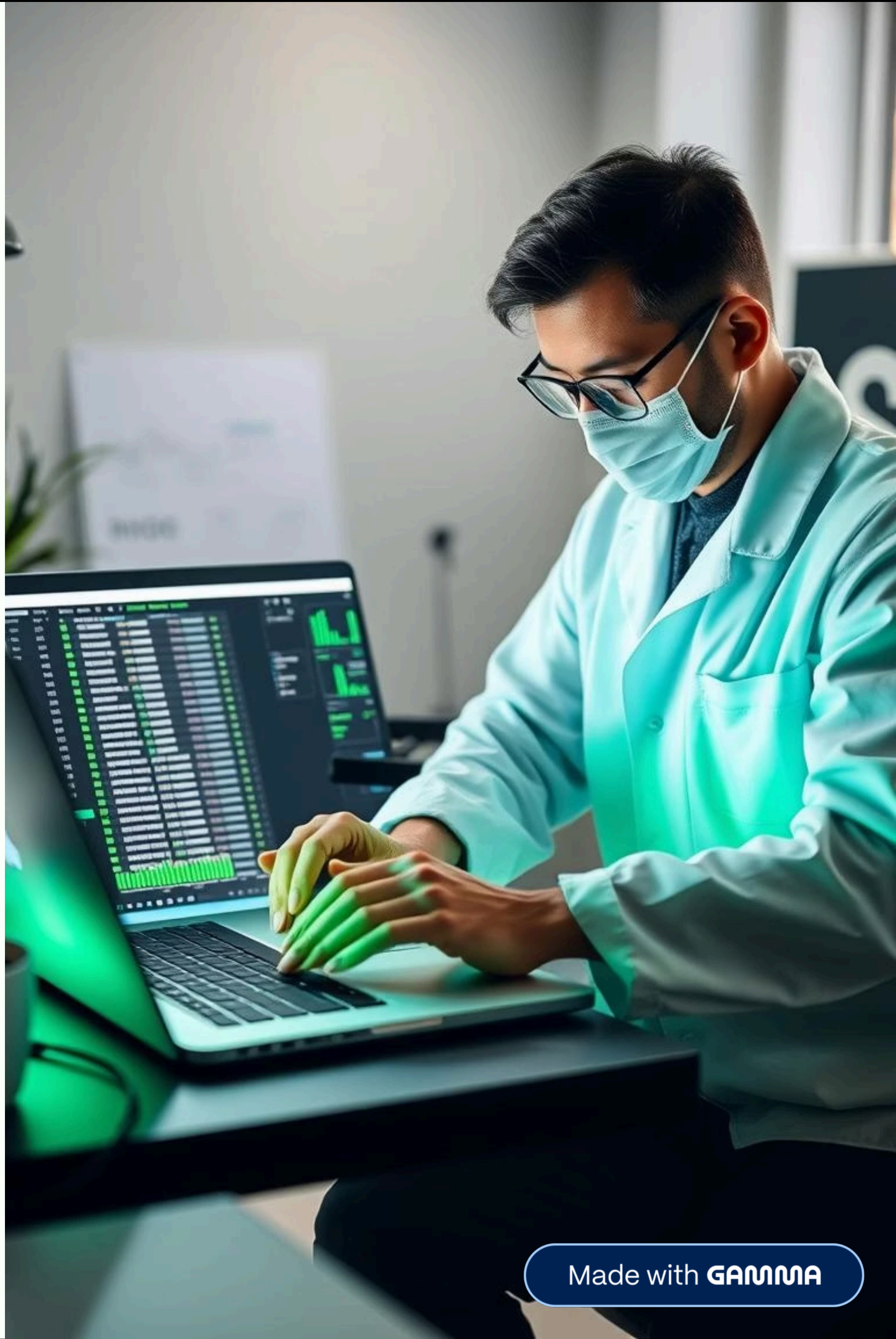
Encoding

Converted categorical variables to numeric format.

3

Scaling

Standardized features for model consistency and improved training.



Modeling and Performance

Models Used

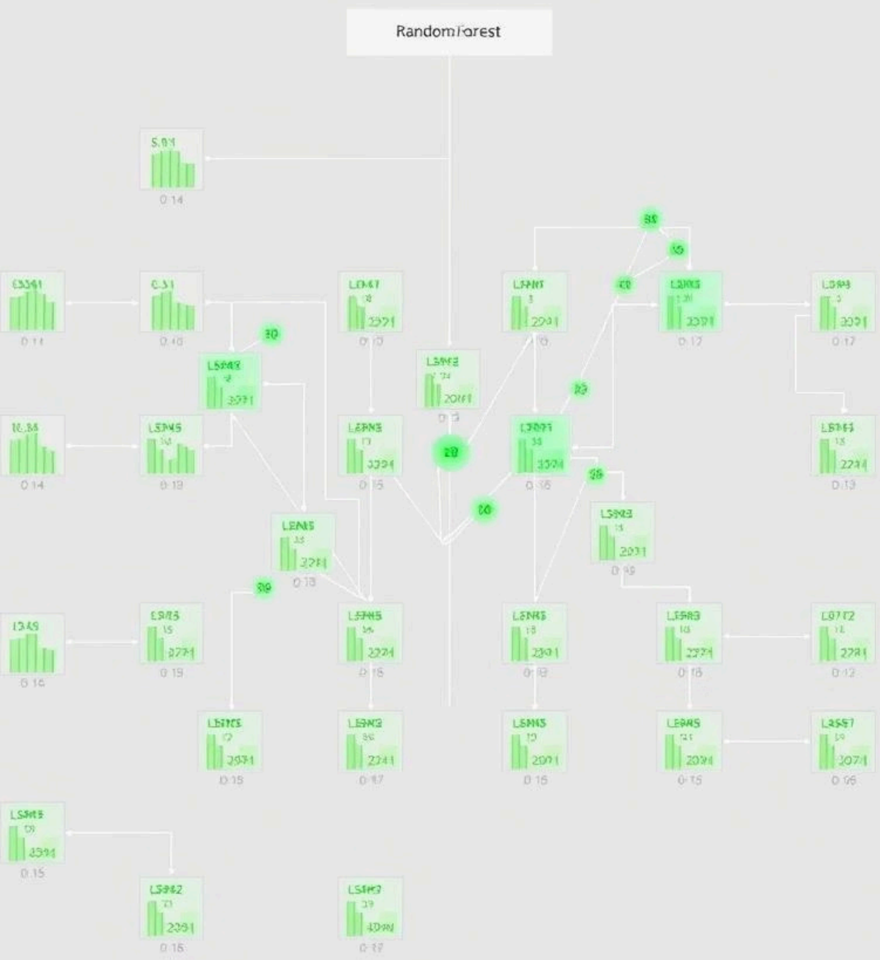
- Random Forest Regressor
- Linear Regression

Performance Metrics

- RMSE: 0.0155
- R^2 Score: 0.9942

Interactive Yield Prediction Example

Feature	Value
Nitrogen	33
Phosphorus	21
Potassium	25
Temperature	29°C
Predicted Yield	0.38 tons/hectare



Key Findings

Model Effectiveness

Random Forest outperformed Linear Regression significantly.

Important Features

- Nitrogen and Rainfall drive yield most strongly.

Conclusion and Recommendations

Summary

Random Forest achieved high accuracy ($R^2 = 0.9942$).

Future Work

- Integrate real-world yield measurements for validation.
- Explore advanced models like XGBoost for improvements.